Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea

and a paragraph describing how that paper supports your design concept.
        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider the project below.
        - Create an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the chempy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.

The goal of my website is to help singers and dancers create a realistic practice schedule based on the activity they want to practice, their skill level, and their available time.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Modacity allows you to create organized vocal practice lists. DanceLog allows you to log your practices. However, there is no app/website that works for both singing and dancing and DanceLog doesn't help you to create the practice schedule, it just lets you log the practice.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The functionality of the program would include personalized schedule generation, a checklist that lets users mark activities as completed, and a progress tracker that records completed practice sessions using browser local storage.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

To create a website to help singers and dancers build personalized practice schedules based on their available time, experience level, then track their completed practice sessions to monitor their progress and consistency over time.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Hobby (singing or dancing), experience level (beginner, intermediate, advanced), available time

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Select activities and distribute practice time.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

A customized practice schedule.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

User, Practice Activity, Practice Schedule, Practice Session, Practice Progress

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Practice Session:
Details I need: Activity type, practice duration, experience level, completion status
Details I will ignore: user's physical appearance, exact practice location, specific equipment they use.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think it could be difficult to figure out how to make the checklist work.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will make sure to type out all of the code given by the LLM rather than copying and pasting it to make sure that the code actually makes sense and that I understand it.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal is to learn more about Drafter to become more comfortable with it.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool to help singing and dancers create a personalized practice schedule to help them monitor their progress over time. What are 3 common problems singers and dancers face when creating practice schedules and what are 3 simple features a basic tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The 3 common problems they said were difficulty managing practice time, lack of structure, and difficulty tracking improvement. They mentioned some of the same prior apps that I did before like Modacity and DanceLog as well as some other ones Andante that tracks practice sessions, daily goals, streaks, and personal reflections and STEEZY Studios that offers online dance classes, guided programs, choreography tutorials, and adjustable video playback. The three features they suggested were a personalized weekly practice planner, a practice goal checklist, and a progress tracker.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The response revised the three website features to have the personalized practice schedule, practice goal setting, and a practice completion checklist. This removed the progress tracker and replaced it with the goal setting.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

This conversation helped me to focus what features my website will have and it helped to make my idea more clear.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck I will use the provided LLM to help me work through it.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I am going to read the error to try to figure out what caused it. I can also use the trace to figure out what went wrong and use that information to try to figure out how to fix it. I could also tell the LLM what went wrong and see if it could fix its mistake.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Can you help me think of a goal for my website? I am thinking of possibly doing a website that has to do with one of my hobbies like dancing or singing.

Absolutely! Dancing and singing could both make really interesting websites, especially because you could create something related to a hobby you actually enjoy.

For your CS1 final project, I would recommend choosing a website that solves a specific problem for dancers or singers, rather than just creating a website that provides information about them.

Since you have about three weeks to build this in Drafter, here are four ideas that I think are realistic and creative.

4 possible website ideas



1. Dance Practice Planner

A website that helps dancers organize their practice sessions and track their progress.

Features:

- Enter dance styles and routines.
- Set weekly practice goals.
- Log completed practice sessions.
- View progress through graphs.

Goal: Help dancers stay organized and consistently improve their skills.



2. Vocal Practice Tracker

A website that helps singers track their singing practice and work toward personal goals.

Features:

- Enter songs they want to learn.
- Track how long they practice.
- Rate their confidence with each song.
- Receive suggested practice activities.

Goal: Help singers structure their practice sessions and monitor their progress over time.



3. Dance Routine Organizer

A website that helps dancers remember choreography and prepare for performances.

Features:

- Create different routines.
- Divide routines into smaller sections.
- Mark sections as learned or needing practice.
- Generate a practice checklist.

Goal: Make learning and remembering choreography more manageable.



4. Personalized Music Practice Generator

A website for people who enjoy both dancing and singing!

Features:

- Select dancing or singing.
- Enter available practice time.
- Choose a skill to improve.
- Generate a personalized practice schedule.

Goal: Help performers make the most of their available practice time by creating customized practice sessions.

My recommendation

I think idea #4, the Personalized Music Practice Generator, would be particularly interesting for your project.

It combines both of your hobbies while having a clear Input-Process-Output (IPO) model, which your assignment specifically asks for.

For example:

IPO Stage	Example
Input	Dancing, 30 minutes, beginner
Processing	Select appropriate activities and divide the available time
Output	A personalized 30-minute practice plan

Your one-sentence project goal could be:

"The goal of my website is to help individuals who enjoy dancing or singing create personalized practice schedules based on their interests, skill levels, and available time."

This would be achievable using basic Python concepts like lists, dictionaries, conditionals, and functions. You wouldn't need an account system, database, or complicated AI integration.

Let's narrow down your idea!

Before we work on your project pitch worksheet, which direction sounds the most exciting to you?

Dance Practice Planner

Vocal Practice Tracker

Dance Routine Organizer

Personalized Music Practice Generator (both hobbies)

I like the hobbies, but want something more creative

We can use your choice to develop a more specific problem, website goal, and set of features without making the project too complicated.

Personalized Music Practice Generator (both hobbies)

Great choice! The Personalized Music Practice Generator combines dancing and singing while solving a real problem: people sometimes want to practice their hobbies but don't know how to organize their time effectively.

Let's develop this into a realistic CS1 final project.

Your website concept

Possible name: Rhythm & Routine

Main goal: Help dancers and singers generate personalized practice schedules based on their interests, experience levels, and available practice time.

How your website would work

Imagine someone visits your website and fills out the following information:

Rhythm & Routine

Example of your website's main feature

Choose your activity

Dance

Singing

Both

Experience level

Beginner

Intermediate

Advanced

Available practice time

15 minutes

30 minutes

45 minutes

60 minutes

Your personalized schedule

Intermediate · Dance · 30 minutes

Warm-up

6 min

Technique

9 min

Choreography

12 min

Reflection

3 min

This is an illustrative prototype; the actual app can round durations to whole minutes.

How this fits your assignment

Your worksheet asks you to identify a problem and explain the Input-Process-Output model.

Here's a possible starting point:

Requirement	Your project
Problem	People struggle to organize effective dance and singing practice sessions.
Input	Hobby, experience level, available time
Processing	Select activities and distribute practice time
Output	A customized practice schedule
Main data	Practice activities, durations, experience levels

Features I recommend including

Start with three main features:

1. Personalized schedule generation.
2. A checklist that lets users mark activities as completed.
3. A progress tracker that records completed practice sessions using browser local storage.

These features would give your website useful functionality without requiring a database or backend.

You could also add a visual progress chart if you have extra time.

One important design decision

Before developing the rest of your project pitch, I think we should decide what makes your

website different from a regular practice planner.

Would you rather your website focus on:

Creating practice schedules

Users receive a personalized plan every time they visit.

Tracking personal improvement

Users can record practice sessions and see their progress over time.

Both!

Users generate personalized schedules AND track their progress.

I would lean toward Option C, provided we keep the progress tracking simple enough for a three-week introductory programming project.

I want option C: Creating schedules and tracking progress. Help me narrow the project scope.

I think Option C is a strong direction! Combining personalized schedules with progress tracking makes your website more useful because users can both plan their practice and see what they've accomplished.

However, since this is a three-week CS1 project, I recommend narrowing it to one main feature and two supporting features.

Your refined project: Rhythm & Routine

Main goal

Create a website that helps dancers and singers build personalized practice schedules based on their available time and experience level, then track their completed practice sessions to monitor their consistency over time.

1. What your website should include

I recommend these three core features:

Feature 1: Practice Schedule Generator

Essential

Users provide:

- Activity: Dancing, singing, or both
- Experience: Beginner, intermediate, or advanced
- Available time: 15, 30, 45, or 60 minutes

Your program generates a practice schedule with appropriate activities and durations.

Feature 2: Practice Checklist

Users can check off activities as they complete them.

For example:

Today's Dance Practice

Warm-up

Technique exercises

Choreography practice

Reflection

Practice completed

0/4

Feature 3: Progress Tracker

Save completed sessions and display:

- Total sessions completed
- Total minutes practiced
- Number of dance versus singing sessions

You could use a simple Matplotlib graph to visualize progress.

2. What I recommend NOT including

This is equally important when narrowing your project.

Feature	Why I'd leave it out
User accounts	Requires authentication and more infrastructure

AI-generated singing feedback	Audio analysis is significantly more complex
Video choreography analysis	Too advanced for an introductory project
Sharing progress with friends	Requires a backend or external service
Music streaming integration	Adds unnecessary API complexity

Your website should focus on generating schedules and recording practice, not analyzing someone's actual singing or dancing ability.

3. Your Input-Process-Output Model

This is especially important because your project worksheet specifically asks for it.

INPUT

User provides their preferences

Activity, experience level, and available practice time.

PROCESSING

Program creates the schedule

Python uses conditional statements and predefined activity lists to select exercises and divide the available time.

OUTPUT

User receives their results

A personalized schedule, an interactive checklist, and an updated progress summary when they save a completed session.

4. A realistic three-week development plan

I would break the project into these milestones:

Week 1 — Build the foundation

Create the homepage, user input form, and basic practice schedule generator.

Week 2 — Add functionality

Implement the practice checklist, completed-session tracking, and browser local storage.

Week 3 — Polish and test

Add a simple progress graph, improve the design, and test different user inputs.

Because Drafter runs in the browser using Pyodide, local storage is a practical way to save progress on the same browser without needing a database.

5. A possible answer for your project worksheet

For Part 1, Question 1, which asks you to identify the problem, you could use this as a starting draft:

People who enjoy dancing and singing may struggle to organize their practice time and consistently track their progress. Without a structured plan, it can be difficult to decide which skills to focus on or recognize how much time has been dedicated to improving. My website aims to solve this problem by helping users create personalized practice schedules and keep track of their completed sessions.

And for your ultimate project goal (Part 4, Question 3):

My ultimate goal is to create an interactive and visually appealing website that combines my personal interests in dancing and singing with the Python programming skills I have learned in CS1. I want to develop a functional application that allows users to create customized

practice schedules, track their progress, and stay motivated to practice consistently.

One final recommendation

Make schedule generation your primary feature and progress tracking your secondary feature.

That way, even if the graph or some of the progress-tracking functionality becomes challenging, you'll still have a complete and usable application to submit.

what would the nouns of the problem be? I think personal improvement would be one

Yes! Personal improvement is definitely an important part of your project. However, for the Data Abstraction section of your worksheet, you want to identify the main *things* your program needs to store information about.

Personal improvement is more of an overall goal, while Practice Progress would be a noun your program can represent using data.

Possible nouns for Rhythm & Routine

I would recommend these five:

1. User: The person using the website to practice dancing or singing.
2. Practice Activity: A specific exercise, such as choreography, vocal warm-ups, or technique practice.
3. Practice Schedule: A personalized plan based on the user's available time and experience level.
4. Practice Session: An individual practice session that the user completes.
5. Practice Progress: A record of completed sessions and the total time spent practicing.

Which noun should you focus on?

For Part 3, Question 2, your worksheet asks you to choose your most important noun and identify 3–4 properties.

I would choose Practice Session, because it connects your schedule generator to your

progress tracker.

For example:

Properties I need	Example
Activity type	Dancing
Practice duration	30 minutes
Experience level	Beginner
Completion status	Completed

Details I would ignore: The user's physical appearance, exact practice location, and the specific equipment they use, since these details aren't necessary for generating a schedule or tracking progress.

Possible worksheet answer

The main nouns of my project are User, Practice Activity, Practice Schedule, Practice Session, and Practice Progress. These represent the important information my website needs to organize personalized practice routines and track personal improvement over time.

One thing to remember: You don't need to track whether someone is actually becoming a better dancer or singer. Your website can measure personal progress through practice consistency, completed activities, and total practice time. This is much more realistic for your CS1 project!

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.